ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice b.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers, addresses and neat ladder diagrams.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 C ✓ 1.1.4 A ✓ 1.1.5 B ✓ 1.1.6 D ✓ 1.1.7 A ✓ 1.1.8 C ✓ 1.1.9 C ✓ 1.1.10 A ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 False ✓ 1.2.7 True ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 G ✓ 1.3.2 J ✓ 1.3.3 B ✓ 1.3.4 E ✓ 1.3.5 H ✓ 1.3.6 I ✓ 1.3.7 F ✓ 1.3.8 C ✓ 1.3.9 D ✓ 1.3.10 A ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 (a) inductive proximity sensor ✓ – detects metal without contact ✓ (b) capacitive sensor ✓ – detects non-metallic materials such as rice through the hopper wall ✓ (c) photo-electric (through-beam or retro-reflective) sensor ✓ – long range, detects any opaque object ✓ (d) reed switch / magnetic cylinder sensor (or limit switch) ✓ – detects the magnet on the piston at the end of stroke ✓ (8)

2.2 NPN (sinking): when on, the output connects the load / input to 0 V ✓✓; PNP (sourcing): when on, the output connects the load / input to +24 V ✓✓. Drawing: brown wire to +24 V, blue wire to 0 V, black output wire to the PLC input; PLC input common to 0 V ✓✓ (6)

2.3 Any THREE ✓✓✓: type of material (steel vs non-ferrous); size and shape of the target; size of the sensor face; temperature; mounting (flush / non-flush); dirt / moisture (3)

[17]

QUESTION 3

3.1 The PLC reads all inputs into memory ✓, executes the program from the first rung to the last using these values ✓, then updates all outputs and repeats the cycle continuously ✓. (3)

3.2 Rung 1: START (NO) in parallel with a seal-in contact (e.g. memory M0.0 or Q0.1) ✓, in series with STOP (NC) ✓, driving a run memory / horn Q0.1 ✓; rung 2: run contact enables on-delay timer T1 with preset 5 s ✓✓; horn Q0.1 on while timer not done ✓; rung 3: T1 done contact ✓ switches on motor Q0.0 (and the horn switches off) ✓ (8)

3.3 Rung 1: sensor I0.2 (NO) ✓ to count input of up-counter C1 with preset 12 ✓; reset input I0.3 ✓; rung 2: C1 done contact (NO) ✓ switches on Q0.2 ✓ (5)

[16]

QUESTION 4

4.1.1 $N_2 = N_1 \times D_1 \div D_2$ ✓ $= 1\,440 \times 120 \div 360$ ✓ $= 480\ r/min$ ✓ (3)

4.1.2 $P = 2\pi NT \div 60$ ✓ $= 2 \times 3{,}142 \times 480 \times 30 \div 60$ ✓ $= 1\,508{,}16\ W\ (1{,}51\ kW)$ ✓ (3)

4.2 Slip: belts can slip, chains do not ✓; noise: belts are quiet, chains noisier ✓; lubrication: belts need none, chains must be lubricated ✓; distance: belts suit longer distances / chains suit medium distances with heavy loads (accept a reasonable comparison) ✓ (4)

4.3 Any THREE causes ✓✓✓: lack of or wrong lubrication; contamination (dirt, water); misalignment; overloading; incorrect fitting; vibration; overheating. Sign (any ONE) ✓: noise / grinding; heat; vibration; play or wobbling of the shaft (4)

4.4 Misalignment causes vibration, extra loads on bearings and seals, wear and coupling failure ✓✓. Method (any ONE) ✓: straight edge and feeler gauge; dial indicator; laser alignment tool (3)

[17]

QUESTION 5

5.1.1 Inductive sensor – detects metal parts ✓✓; photo-electric / capacitive sensor – detects any part arriving at the pusher ✓; cylinder position (reed) sensors – confirm the cylinder is extended / retracted ✓ (4)

5.1.2 Inputs (any THREE) ✓✓✓: START, STOP / emergency stop, part sensor, inductive (metal) sensor, cylinder extended / retracted sensors. Outputs (any TWO) ✓✓: conveyor motor, solenoid valve of the cylinder, indicator / alarm light (5)

5.1.3 Any SIX logical steps ✓✓✓✓✓✓: press START – conveyor runs; part sensor detects a part; inductive sensor checks whether it is metal; if metal, wait until the part is in front of the cylinder (timer or position sensor); energise the solenoid valve – cylinder extends and pushes the part into the bin; extended sensor confirms, valve de-energises and the cylinder retracts; plastic parts are ignored and travel to the end; STOP or emergency stop halts the conveyor (6)

5.1.4 Any TWO ✓✓: emergency stop; guards over moving parts and the cylinder; pneumatic pressure relief / dump valve; warning signs or lights; lock-out facility (2)

5.1.5 Conveyor (any ONE) ✓: check belt tension and tracking; lubricate bearings; clean rollers. Pneumatics (any ONE) ✓: drain water from the filter; check for air leaks; check the lubricator oil level / pressure setting. Correct pairing ✓ (3)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE